

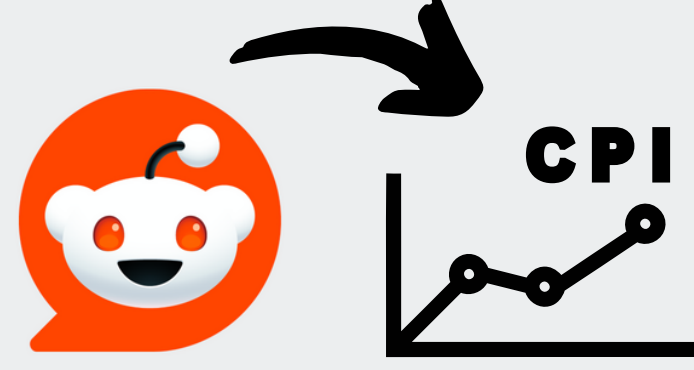
Constructing a Reddit-based Inflation Index

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01 PROBLEM STATEMENT

- Official inflation indicators, such as the Consumer Price Index (CPI), are **published with a time lag**, limiting their usefulness for **real-time monitoring** and decision-making.
- At the same time, millions of **Reddit users publicly share experiences** related to price increases, cost-of living concerns, or financial hardship.
- This project investigates whether **Reddit can serve as a meaningful data source** for reconstructing the U.S. inflation index. Using historical Reddit data from 2012 to 2024, we evaluate whether **natural language processing models can approximate official CPI trends**.
- If successful, this approach would demonstrate the feasibility of using **social media as a real-time complement** to traditional economic indicators.



02 DATA

- 3.1 TB** Reddit submission dumps (2005–2024) from **Academic Torrents**
- 2012 - 2024** quarterly months (03, 06, 09, 12)
- Subreddits filtered** by inflation relevance (e.g., "food", "economy")
- At last **3 words**, English
- Post score ≥ 1**
- Must contain **≥ 1 inflation-related keyword** (e.g., "price increase", "groceries", "rent")

SUBREDDITS	Vol.
r/food	19,609
r/personalfinanc	13,918
r/frugal	9,236
r/economy	9,236
r/povertyfinanc	13,19
r/Costco	5,973
r/AskAnAmerica	4,635
r/walmart	3,889

03 MODELS

03.1 VADER

- Lexicon-based sentiment model**, designed for short social media texts
- Computes a compound score $\in [-1, 1]$ **based on word valence**
- For each Reddit post, **title + text were merged** and scored
- Formular:**
 - $\text{compound} = \frac{x}{\sqrt{x^2 + \alpha}}$
 - x : Valence values
 - α : smoothing constant for normalization
- $\text{avg_sentiment}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{compound}_i$
- Example:**
- "I'm extremely happy with the lower food prices!":
 - happy value: 3.2
 - extremely: intensifier (+29.3%)
 - $x = 3.2 \times (1 + 0.293) = 4.14$
 - $\text{compound} = \frac{4.14}{\sqrt{4.14^2 + 15}} \approx 0.73$

03.2 BERT (Score Supervision)

- Transformer-based** sentiment model using **DistilBERT**
- Reddit post **scores (upvotes/downvotes)** used as **proxy labels**
- Thresholds:**
 - Score $\geq 3 \rightarrow$ positive
 - Score $\leq 2 \rightarrow$ negative
- Training:** Learns linguistic patterns typical for highly rated inflation complaints
- Application:** Model predicts if a new post resembles a "positive" pattern $\rightarrow$ monthly average = sentiment index
- Formular:**

$$\text{SentimentIndex}_m = \frac{1}{N_m} \sum_i = 1^{N_m} \hat{y}_i$$
- Examples:**
- "My grocery bill has doubled in the last two months, I can't keep up."
 - Score: 42 $\rightarrow$ positive (high resonance)
- "Rent is fine if you just budget better."
 - Score: 1 $\rightarrow$ negative (low resonance)

03.3 BERT (CPI Regression)

- Regression-based** model, trained to **predict CPI values directly**
- For each month:** Reddit posts $\rightarrow$ aggregated text $\rightarrow$ mapped to official CPI
- Input:** aggregated monthly text X_m
- Target:** normalized CPI $y_m^{(norm)}$
- Formular:**

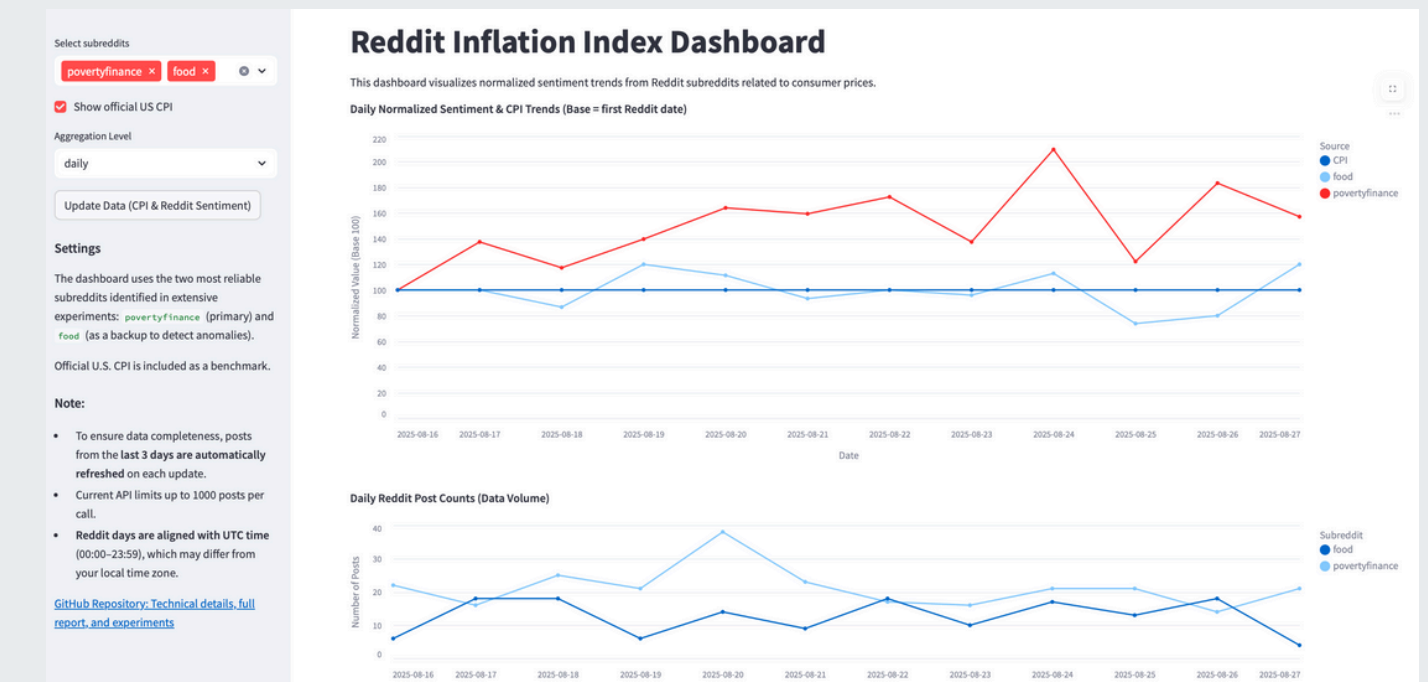
$$\hat{y}_m = f\theta(X_m) \quad \text{and} \quad y_m^{(norm)} = \frac{y_m - \mu}{\sigma}$$
- Trained using **DistilBERT** with MSE loss
- Learns to **associate economic language with actual inflation levels**
- Examples:**
 - "My rent increased by 30% this year and groceries are insane." $\rightarrow$ Indicates high inflation ($\uparrow$ predicted CPI)
 - "Finally seeing gas prices go down, things might be normalizing." $\rightarrow$ Indicates low inflation ($\downarrow$ predicted CPI)

METHODOLOGY

- Staged workflow:** Initial evaluation on a **small development set** (Dec 2016–2024) to identify top-performing subreddits.
- Resource-efficient:** Only **best 4** subreddits used for full VADER analysis; best 2 selected for BERT-based models.
- Temporal smoothing:** Quarterly predictions **aggregated** to semiannual and annual levels to reduce volatility.
- Goal:** Maximize model performance while **minimizing computational cost** and noise sensitivity.

05 LIVE-DASHBOARD

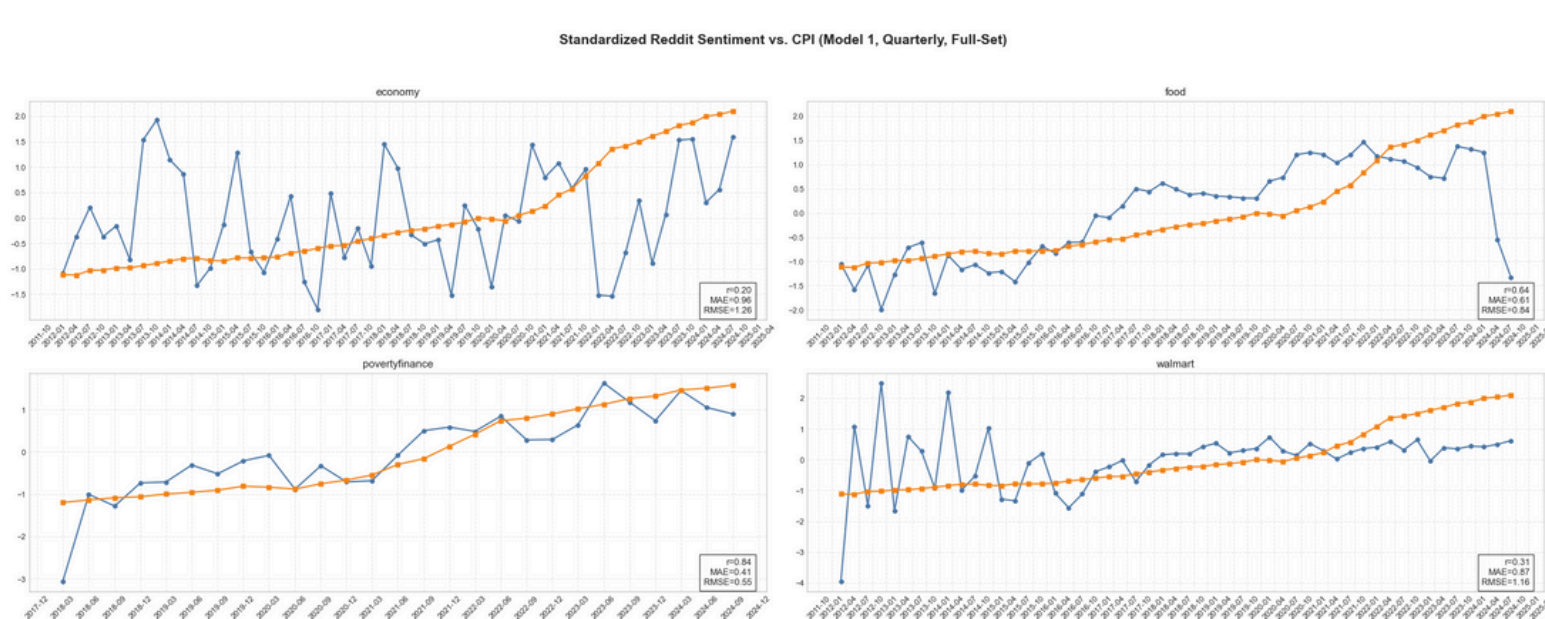
- Real-time Reddit monitoring (Streamlit):**
 - Daily, weekly, and monthly** sentiment trends visualized using beste performing Model (BERT Score Supervision)
 - New Reddit data is fetched and reprocessed on each refresh; CPI updates included
- Transparency & context:**
 - Post volume and top-rated examples shown; dashboard reflects same filtering and model setup as offline analysis



04 RESULTS

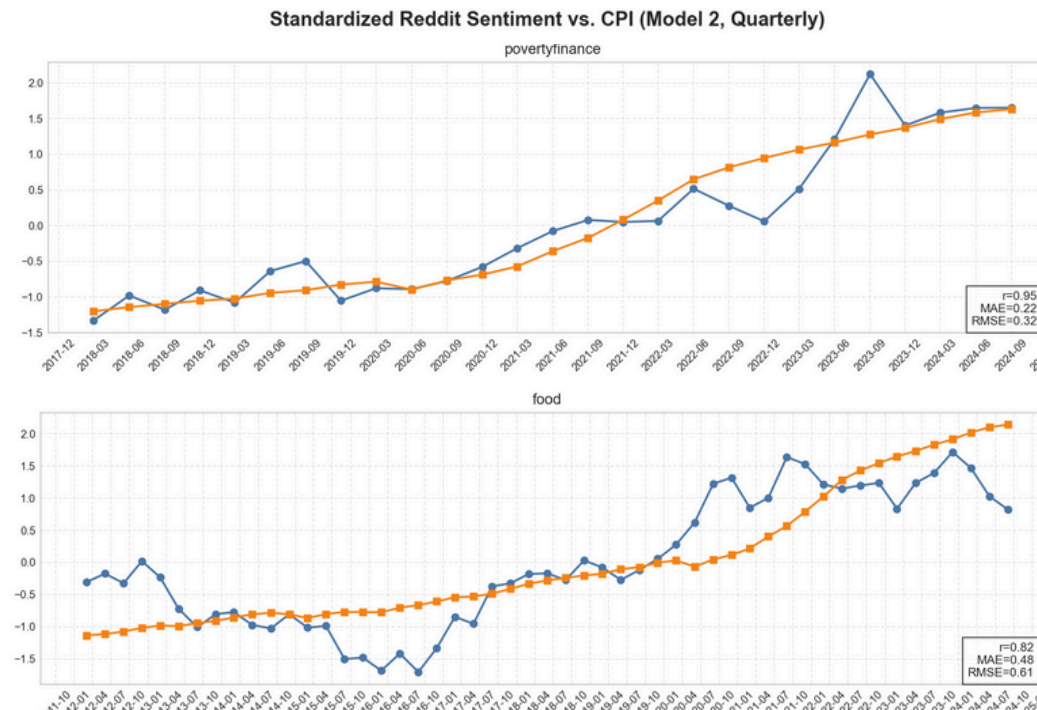
BERT (Score Supervision) clearly **outperforms** the other models across both subreddits. **Especially for povertyfinance**, the signal aligns almost perfectly with CPI fluctuations, even without direct inflation labels.

04.1 VADER



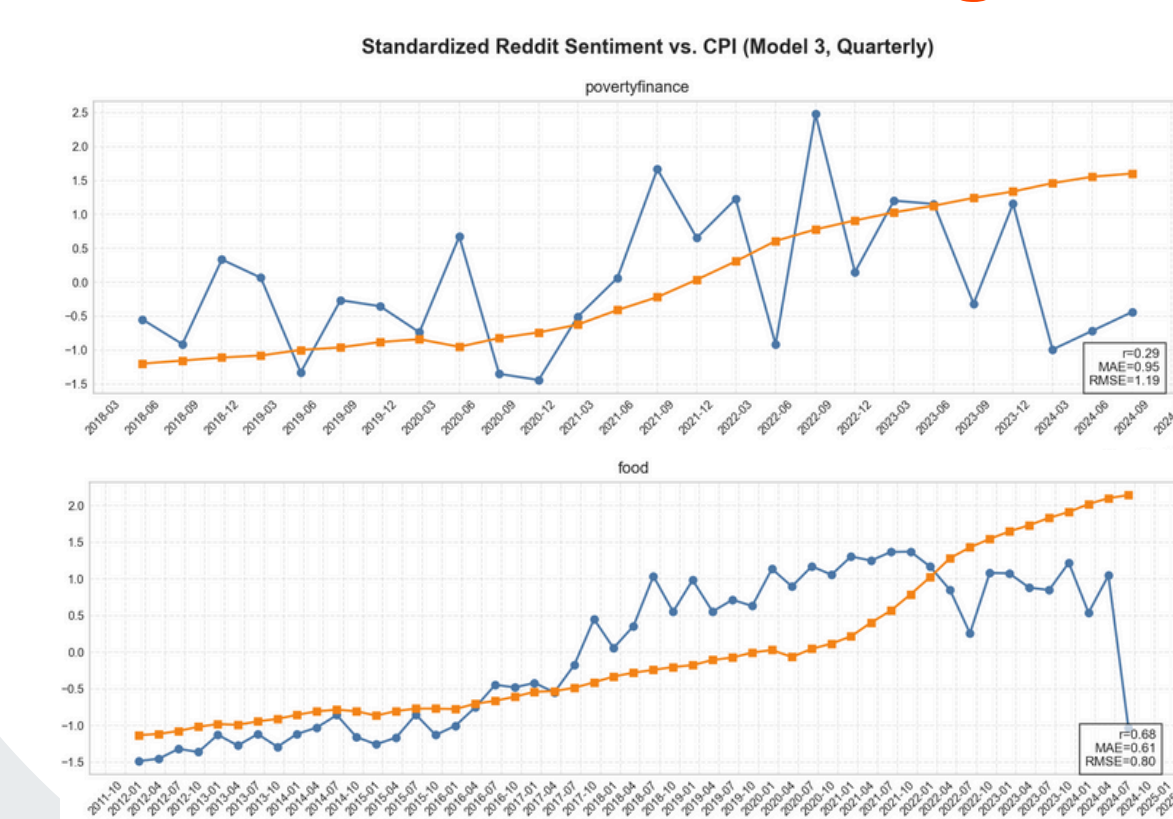
Povertyfinance shows strong correlation ($r = 0.84$), but **others more noisy** (e.g., economy $r = 0.20$)

04.2 BERT (Score Supervision)



highest accuracy with povertyfinance ($r = 0.95$, MAE = 0.22), robust trend matching

04.3 BERT (CPI Regression)



mixed results; food improves over VADER ($r = 0.68$), povertyfinance drops sharply ($r = 0.29$)